

Architectural Particular Specification

BALUSTRADE AND METAL RAILING

SECTION 10.5

Rev	Date	Description
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SECTION 10.5

BALUSTRADE AND METAL RAILING

PART 1 GENERAL

1.01 SUMMARY OF WORK

- A. All works under this section shall be executed by the Contractor in accordance with the drawings:-
- B. The scope of works shall be defined by the architect's drawings, articles of the specifications and other documents and instructions which pertain to the main contract works.
- C. The Main contractor shall design, engineer, test, fabricate, supply, deliver, install, and guarantee all construction necessary to provide a complete system of Balustrade / Railing including any measures that may be required to that end, notwithstanding any omissions or inadequacies of drawings and/or specifications.
Tender sum to be itemized and broken down, with rates entered in the Schedule of Rates. The prices therein will be used as the basis for pricing any additional or omitted variation works.
- D. The main contract works shall include, but not limited to:-
 - 1. GMS or stainless steel posts, mullions and transoms, including capping to transom and / or mullion members, any metal infill bars, metal mesh or other components as shown on detail drawings.
 - 2. Other steel reinforcement required for a complete installation (whether these structural frame reinforcement are illustrated on drawings or not) and a preparatory work to the adjoining structure for attachment of the works.
 - 3. Any Structural Glass and Glazing Materials
 - 4. Fixings, connections and provision of anchorages and/or cast-in embedments in concrete. All Aluminium / Steel Supporting Frames should be anchored to the concrete structure.
 - 5. Glazing accessories including but not limited to glass spacers, gaskets, sealants (including structural sealant), backer rod, Norton tape, bi-metallic corrosion insulation/isolation tape/rod, etc.
 - 6. Ironmongery and fittings, if any.
 - 7. Sills, copings, water bars, flashings, seals and other attachments, if any.
 - 8. Isolation of dissimilar metals and moving parts.
 - 9. Sealants completed with appropriate accessories such as backer rod and other filling materials, etc. within work of this section and at boundaries / interfaces with work of other sections.
 - 10. Flashing and sealing of surround to GMS frames by approved waterproof grout and slurry coat, if any.
 - 11. All essential materials/drawings/calculations for submissions to the relevant authorities for obtaining essential approval and/or consent for commencement of field works. This should include submissions to the BD.
 - 12. Test mock-up and corresponding tests (including performance testing as required by Buildings Department and other relevant authorities) carried out by an independent laboratory approved by the Architect and to the satisfaction of relevant government authorities.

13. Any material tests as required by the local authorities or the manufacturer to confirm the quality & workmanship of the installed materials/works
 14. Final cleaning and appropriate protection to the installed balustrade/railing system against accidental damages during the course of construction works and prior to handover to the Employer all to the satisfaction of the Architect.
Proposed method and means of protection to be submitted for Architect's Approval.
 15. Lightning protection, equipotential bonding and means for connection to earthing terminals of the development. The design of the components, assembly, and operation system of the roller shutters, including all motor equipments, gears, boxes, hoods, guides, and other parts shall be the responsibility of the Contractor.
- E. Apart from the above, the scope of works shall also include, without being limited to:-
1. All cover panels and cappings where indicated.
 2. All caulking, sealing & flashing including sealing, backing rods and joint filler, etc. at junction with roof waterproofing, external wall tiling, external masonry works & other areas in contact with the external concrete wall.
 3. Continuity of various metal parts (whether fixed or movable) to facilitate earth bonding and link up of lightning system.
 4. All bolts, angles, fasteners, anchors, inserts and channels in concrete structure for the anchorage of all works under this section.
- F. The main contractor shall also be responsible for:-
1. The design, engineering fabrication, supply, transportation to site, installation and testings of the works under main contract including grouting behind frames, caulking, glazing, pointing etc. in accordance with the tender documents.

Save where explicitly excluded, the main contractor is to provide all components, fixings and accessories etc. required to complete the system as is intended, regardless of whether such have been explicitly described or not.
 2. Shop drawings, engineering proposal, structural calculations, test reports endorsed by an independent testing laboratory upon completion of performance tests as required by Buildings Department, structural sealant compatibility tests to be performed by an independent laboratory to be approved by the Architect / relevant authorities, impact load tests to tempered glazings to demonstrate their capability in withstanding the designed wind load and / or the horizontal imposed loads as referred to in Table 3 of Building (Construction) Regulations, and all other relevant documents and data of the GMS frame, sealant, and glazing / infill panel / mesh to be endorsed by a Registered Structural Engineer acceptable to the Architect / Engineer for submission to Buildings Department for obtaining necessary approvals and consent for his works. The contractor shall bear the consequences of any delay due to the failure in obtaining the necessary government approvals and consent.
 3. Scheduling and monitoring of the work,
 4. Samples, full size unit mock-ups and test units,
 5. Performance testing of glass balustrade/metal railing assembly together with corresponding test reports endorsed by an independent testing laboratory to be approved by the Architect;
 6. Co-ordination work or other builder's works and other trades with work interfacing.
 7. Protection of materials and works, handing-over to the Employer all to the satisfaction of the Architect. Proposed method / means of protection to be submitted for approval of the Architect.
 8. Warranties,

9. All final exterior and interior cleaning of the Glass Balustrade / Railing by Main Contractor.
 10. All hoisting and temporary site services for proper execution of the works.
 11. Cost of all delivery & hoisting of material.
 12. Design, supply and install the earthing / equipotential bonding in accordance with I.E.E. regulations and provide a termination point for the Electrical Sub-contractor to connect and integrate it into that of the rest of the building.
 13. Make good and replace any damaged parts.
 14. Provide spares and tools prior to Practical Completion of the Main Contract.
- G. This specification and the architectural drawings showing balustrades / metal railings are for the sole purpose of defining the design intent and performance requirements only. To avoid any misunderstanding or lack of interpretation, the contractor is hereby advised that the responsibility for the works is totally his, including but not limited to the design, structural calculations, waterproofing performances, carrying out of performance testing and preparation of corresponding reports to be endorsed by an independent testing laboratory acceptable to relevant government authorities and the Architect / Engineer, shop drawings, material installation, warranties, and related documentation and certification as required by Buildings Department and other relevant authorities, and those stipulated under the Buildings Ordinance (Cap. 123), its subsidiary legislations, and/or other enactments as stipulated in this specification.
- H. Schedule fabrication, delivery of bolts, anchors, brackets, holes, slots, etc. and installation including insertion in concrete structures with works of other Sub-contractors when and as required to meet the construction programme. Co-ordinate sizes of fabricated glass balustrade with glazing size and supporting frame size.
- I. Manufacturers and erectors shall have proven achievement and experience in similar work for at least ten years and are subject to endorsement by the Architect.
- J. Other related works, but is not limited to the following:-
1. Reinforced concrete structure and external tiling works;
 2. External and Internal Stone Installation works / finishing;
 3. Floor coverings and ceiling construction;
 4. Mechanical and electrical systems (except lightning protection and equipotential bonding system);
 5. Waterproofing;
 6. Roofing and roof finishes;
 7. External paving & asphalt works;
 8. Fixing of anchorages into structure.

1.02 REFERENCE

(NOT USED)

1.03 QUALITY ASSURANCE

(NOT USED)

1.04 PERFORMANCE REQUIREMENTS

- A. Only the latest editions of the referenced codes and standards as of date of tender shall apply.
- B. The standards, codes, and specifications referenced below and elsewhere in these specifications shall apply; however, similar codes or standards used in the home country of the manufacturer may be used if applicable, subject to acceptance of local authorities. In such case the manufacturer shall submit documentation satisfactory to the Architect that such code is essentially equal to the one specified.
1. Specifications for the Design Fabrication and Erection of Structural steel for Building issued by the American Institute of Steel Construction (AISC).
 2. The Code for Welding in Building Construction issued by the American Welding Society.
 3. Specification for Structure of Aluminium Alloys issued by Aluminium Association (AA).
 4. Stainless Steel Designation issued by American Iron and Steel Institute (AISI).
 5. The specified documents of the American Society for Testing and Materials (ASTM).
 6. The specified documents of the International Organisation for Standardization (ISO).
 7. Japan Industrial Standard (JIS).
 8. Specification for Metal Window Wall by the American Aluminium Manufacturers Association (AAMA).
 9. Specification for Metal Window Wall by Japan Window Wall Manufacturers. (JCWA).
 10. All current legislations, regulations, Code of Practice and Practice Notes for Authorised Persons / Registered Structural Registered Contractors under the Buildings Ordinance (Cap. 123), Buildings Department, Hong Kong.

The contractor should strictly observe and comply with the requirements laid down in PNAP 53, 59, 106, 130, 135, 160, 186, 235, 239, 248 & 256 (which shall not be deemed exhaustive) in the course of their design, fabrication manufacturing, drawings and documents submission preparation, quality control (both off- and on-site), actual construction on-site, testing / commissioning, maintenance during Defects Liability and Warranty Period, and record plan / documents submission preparation for facilitating the application of Occupation Permit upon works completion for all works under this contract.
 11. British Standard specification (B.S. or B.S.S.)
 12. Standards and Code of Practice for the design of structural steelwork elements shall also refer to the Specification for Structural Steel Works. Fire Resistant Shutter
- C. Unless otherwise specified, all materials shall be of the best quality of the type specified and consistent in colour finish.
- D. Materials and methods of engineering development other than indicated or implied by the drawings and specification may be employed when such materials and methods conform to all of the following; the design intent; the design requirements; the applicable codes and standard; they are endorsed by the Architect.

1.05 ENVIRONMENTAL REQUIREMENTS

(NOT USED)

1.06 ENVIRONMENTAL, CLIMATIC & PSYCHOMETRIC CONDITIONS

(NOT USED)

1.07 SUBMITTALS

A. Submission Schedule

1. Immediately after award of the contract, the contractor shall submit a complete submission schedule (in a format to be approved by the Architect) to the Architect for approval listing all the items requiring approval from the Architect, their programme of submission and expected date for getting approval from Architect for their ordering and fabrication of materials. The list of items shall include the followings:-

- a. Samples of materials,
- b. Shop drawings,
- c. Technical data, technical information & catalogue of materials & equipment,
- d. Mock-up,
- e. Structural calculations.

The above submission schedule will be treated as an indication of the contractor's intended programme for submission & application for approval only.

2. Sub-standard Submission Materials

The contractor should note that no consequential claim on time and monetary compensation from late approval of Architect on any submission materials required under this specification will be entertained should such late approval be caused by the enduring poor quality or sub-standard submission materials.

B. Shop Drawings

1. The Contractor is required to submit adequate sets of complete scaled and dimensioned elevation / sectional drawings and full size details of all parts of the main contract works for Architect's approval. Work to be done by other trades shall be indicated. Full specifications of the materials offered, including product name and location on drawings shall be furnished. It is the responsibility of the main contractor to verify at the building site all dimensions shown on Architect's drawings. Six copies and one reproducible print for each of the final approved drawings shall be submitted for the Architect's and Owner's use.
2. The main contractor shall provide drawings showing all building materials in place and drawings shall include sufficient elevations, floor plans, sections, full size details and explanatory notes that fully substantiate the various systems under this main contract. Different types of elements and the like must be identified clearly on the elevation drawings. The types, dimensions, and sectional properties of all structural elements, temporary or permanent fixings, weldings, protection to metals and dissimilar metals must be identified clearly in all details & sections. Details shall be fully drawn (not outlined).
3. In particular, drawings shall include but not limited to the following information:
 - a. Joinery and internal seals;
 - b. Glass and metal thickness in particular fixing lugs and ties etc.;
 - c. Metal alloy, temper and finish;
 - d. Glass strength, tint, coating, opacity, weight and safety backing.
 - e. Fastener alloy, plating, diameter, length, and spacing;

- f. Glazing materials identification and minimum dimensions of structural silicone glazing, where used;
- g. Sealant identification by product name;
- h. Relative layout of walls, beams, columns and slabs with dimensions noted;
- i. Dimensioned position of glass edge relative to metal daylight;
- j. Provisions for thermal movements and building movements;
- k. Location of, and details for, any embedded anchors.
- l. Connection and interfacing detail with adjacent / adjoining structures, walls, beams, M/E accessories, etc. and all interior / exterior finishing materials.
- m. Earthing provision.
- n. Any temporary fixings for the works during installation.
- o. Instructions and explanatory details for the sequence of installation and the installation of all materials, including the glass and re-glazing procedure.

C. Structural Calculation and Submission Drawings

The main contractor shall:-

1. Furnish structural calculation and relevant government submission drawings verified and endorsed by a Registered Structural Engineer approved by the Architect / Engineer, in connection with the design of the main contract works, which are in compliance with all applicable Hong Kong Codes and other statutory enactments.

The drawings shall include details of the materials, method of fixing, structure details including erection procedure and calculations to show that glass balustrade / railing are of adequate strength and stability.
2. The design calculation shall include a synopsis stating the relevant codes and standards, the design loadings and the design consideration, and relevant calculations that substantiate all glass balustrade / metal railing systems under this main contract, and their connection and anchorage satisfy the requirements of strength and stiffness under both the working and the test load conditions.
3. The main contractor shall prepare calculations to satisfy the wind load indicated for the glass balustrade under this main contract and in accordance with good and practical theory and current design practice and in accordance with the rules of the Aluminium Association, AISC, AISI, ACI, Building Ordinance (Cap. 123), its subsidiary regulations, codes of practice and other relevant enactments, including analysis for wind and dead loads on framing members, anchors and concrete inserts. Show section property computations for framing members and submit full-size drawings.
4. These structural calculations and submission drawings must be prepared specific for this project works. Existing test reports, data from other projects, or untried or unproven theories shall not be acceptable. Where a conflict between the various codes, legislations, specifications or agencies exists, that code, legislation, specifications, or agencies which has the most conservative requirements shall govern.
5. The verified and endorsed, structural calculations shall be submitted for checking, endorsement by the Registered Structural Engineer of the project for submission to Buildings Department. The checking and endorsement by the Project RSE shall not relieve the responsibility of the contractor in obtaining on time approval and consent for the works.

D. Method Statement and Detailed Programme

1. The contractor shall submit a comprehensive method statement and a detailed programme for the installation to the Architect. The programme shall show

various stages of design, shop-drawings and structural calculations preparation / submission, submission of drawings / calculations to Buildings Department, carrying out of performance testings, sampling, testing, fabrication, delivery, mock-up preparation, installation of the works, etc.

2. The method statement and the programme shall be the result of co-ordination with other relevant Sub-contractors. The document shall also be incorporated into the master programme, process and later assessment of claims.
3. These documents may be modified from time to time to suit site conditions and shall be used to monitor progress and quality.
4. Should the contractor consider it necessary to fabricate in bulk before endorsement in order to meet completion date, it is entirely at his own risk and no compensation in time or cost will be granted if the material and / or workmanship is eventually not acceptable to the Architect.

E. Samples

1. Submit for approval 3 sets of labelled samples of each required type and colour of metal finish on 300 x 300mm aluminium sheets. Samples must show the extremes of colour and texture variations. Samples will be reviewed by the Architect for colour and texture only.
2. Submit for approval 3 sets of labelled samples of each required types of glass. Samples to be of the required colour, thickness and photometrical performance as specified; size of samples to be 600 x 600mm. Samples must show extremes of:
 - a. Light transmission;
 - b. Reflectance;
 - c. Colour of transmitted light as viewed on a white surface, and
 - d. Mottling of coating for glass spandrel.One set is defined as 2 pieces of each of the 4 properties. One of the piece to demonstrate the high extremes and one to demonstrate the low extremes.
3. Submit for review 3 sets of samples of (where applicable samples shall be 300mm in length):-
 - a. Glazing sealants; (including structural sealants).
 - b. Tapes;
 - c. Setting blocks;
 - d. Side blocks;
 - e. Gaskets;
 - f. Shims;
 - g. Ironmongery;
 - h. Sweep and compression seals;
 - i. Flashing or any other waterproof devices;
 - j. GMS sections for the various supporting systems.
 - k. Brackets, channels, bolts and other fixing systems. Colour chart should also be submitted for those elements with ranges of colour available, such as sealant and gasket, for Architect's selection.
4. All samples are to be clearly labelled and identified. Full technical literature and specifications must be provided with the samples.
5. Approved samples shall be retained by the architect, for comparison with production materials.

6. Samples should show fabrication techniques and workmanship of component parts, and the design of accessories and other concealed or exposed auxiliary items before fabrication of the work proceeds.

F. **Maintenance Particulars**

The contractor shall submit a maintenance manual for the glass balustrade / railing inclusive of all metal parts, glass and organic coated finish.

G. **Sub-contractors and Suppliers**

The contractor shall submit a full list of all specialists and suppliers that are proposed to be associated with the works within one month from the award of the contract.

- H. Upon completion of work, contractor shall supply the approved OPERATION AND MAINTENANCE MANUAL (instead of “two instruction manual” stated in ASD GS 2012 clause 17.44) including all circuit diagrams, schematic diagrams, approved equipment shop drawing, schedule of component parts, priced list of recommended spares, and other technical information provided by the manufacturer. A draft of the manual shall be submitted in advance to the Architect for review and approval before final submission.

1.08 PRE-INSTALLATION CONFERENCE

(NOT USED)

1.09 DELIVERY STORAGE AND HANDLING

A. **Planning of Delivery**

The contractor shall submit preliminary planning reports of the whole stage of the delivery from the factory to the site paying careful attention to the traffic and site condition. The planning shall include packing, storage and unpacking.

B. **Planning of Erection**

The contractor shall include in the above planning a description of the procedure of delivery, hoisting, storage, fixing, scaffolding, temporary working stage or gondola, protection and cleaning.

- C. Package and store materials in a manner that will prevent surface damage or contamination, distortion, breakage of structural weakening.

- D. Replace any materials damage during manufacture, shipping, storage or erection.

- E. Protect materials in place from contamination and damage.

1.10 MOCK-UPS

This part shall be read in conjunction with Clause 8.03 of Specification Preliminaries for the requirements for prototypes and trial areas.

1.11 TESTING

(NOT USED)

1.12 WARRANTY

- A. This part shall be read in conjunction with Clause 8.04 of Specification Preliminaries for the list of material/ system requiring joint written guarantees.
- B. The main contractor shall provide a written warranty agreeing to repair or replace defective materials and workmanship during the warranty period.
- C. Defective materials and workmanship include, but are not limited to:-
1. Any abnormal deterioration, deformation, clippings, aging, rusting, corrosion or weathering of the work.
 2. Water leakage under conditions equivalent to, or less severe than, those specified.
 3. Air leakage exceeding specified limits.
 4. Failure of operating parts to function normally.
 5. Structural failure due to pressures and forces up to specified limits.
 6. Sealant (including structural silicone) loss of adhesion, loss of cohesion, cracking or discoloration.
 7. Disengagement of gaskets or weather-strips.
 8. Deterioration or discoloration of aluminium and cladding finish in excess of normal weathering and aging.
 9. Glass breakage due to pressures up to specified values, thermal stress, defective glass or damaged (prior to or during construction) glass, and resultant damage and breakage caused by falling glass.
 10. Spontaneous breakage of tempered glass.
 11. Delamination or discoloration of laminated glass or panels.
 12. Loss of effective glass bite due to shifting of glass.
 13. Loss of effective glass bearing on setting blocks due to shifting of glass and/or blocks.
 14. Collapse of insulation.
 15. Failure of work to fulfill other specified performance requirement.
 16. Cracking, peeling or discoloration of glass reflective coating.
- D. The warranty does not include damage caused by vandalism, or natural conditions exceeding the performance requirements. However, the warranty does include failures or defects for which the causes cannot be determined. The warranty and its enforcement shall not deprive the Employer of other action, right or remedy available to him.
- E. The warranty period for the balustrade system / railing system shall be **ten (10)** years from the certified date shown on the Certificate of Substantial Completion of the Main Contract issued by Architect, unless otherwise stated. The system warranty includes all materials, plant and labour. All costs and expenses arising out of or in connection with the obligation herein shall be solely borne by the main contractor.
- F. The main contractor shall review and inspect the work of other Sub-contractors and resolve and report to the architect any discrepancies or problems noted which could result in a warranty coverage conflict should any failures or problems occur. In the event of any disagreements between Sub-contractors relative to the system performance and warranty coverage, the final decision concerning which party shall be responsible shall be made by the owner, Architect and/or the consultant.

- G. The warranty shall include, but not limited to, the following:-
1. A reflective glass product whose reflective coating cracks, peels or discolors shall be replaced at no charge during the warranty period.
 2. Laminated glass which delaminates shall be replaced at no charge during the warranty period as specified above.
 3. Colour and finish on aluminium which cracks, peels, fades in excess of specified limits or chalks in excess of specified limits shall be replaced at no charge during the warranty period as specified above.
 4. Defects in structural anchor, stainless steel finishes and aluminium anodizing and powder coating / PVF2 coating shall be replaced at no charge during the warranty period as specified above.
 5. Failure to provide warranties that meet the above stated requirements may lead to disapproval of the proposed glass and materials and the contractor shall re-submit glass and materials from alternative services that are able to provide warranties that meet such requirements.
- H. All warranty is to include for the cost of removal and replacement of all defective work at no extra cost to the Employer at such time as designated by the Employer during the warranty period.

1.13 SPARE PARTS

This part shall be read in conjunction with Clause 8.05 of Specification Preliminaries for the list of spares and spare parts.

PART 2 PRODUCT

2.01 APPROVED MANUFACTURER

(NOT USED)

2.02 MATERIALS

A. Measurements

All true measurements as shown on the tender drawings are in the metric system. The main contractor shall be responsible for verifying all the dimensions and actual conditions on site.

2.03 ANCILLARY MATERIALS

(NOT USED)

PART 3 EXECUTION

3.01 EXAMINATION / INSPECTION

- A. Performance Testing Particulars
The contractor shall submit a procedure plan for the performance test of the various systems under this contract.
- B. Performance Test Report
The contractor shall submit 6 copies of reports including photographs of the performance test for Glass Balustrade required in this specification.
- C. Planning of Inspection of Component
The contractor shall submit preliminary schedule of material specification and procedure for inspection of the quality of components of the Glass Balustrade/Railing under the contract during the fabrication in the plant.
- D. Report on Inspection of Components
The contractor shall report on the results of the inspection of the components in accordance with the "Planning of inspection of components" mentioned in above paragraph C.
- E. Inspection of Erection Particulars
The contractor shall submit a preliminary schedule on the procedure for inspection during installation so as to maintain quality control on the site.
- F. Report on Inspection of Erection
The contractor shall submit a report on the inspection of the erection in accordance with the above paragraph P.
- G. Record Drawing and Photographs
 - 1. Before the date of completion of the work the contractor shall prepare and supply to the Architect, for forwarding to the Employer, Record Drawings of the whole of the work as installed and Record Photographs as requested by the Architect.
 - 2. These drawings and photographs shall be specially prepared for record purposes and the copies as finally provided to the Architect shall consist of two(2) sets of reduced size prints together with soft copy of the drawings and two(2) sets of photographs.
 - 3. Statement of Application
The contractor shall submit a written statement verifying that all works, including framing, sealants, glass and glazing components, complies with the requirements of the contract documents and that installation methods are adequate and proper for the conditions of installation and use.

3.02 PREPARATION

(NOT USED)

3.03 INSTALLATION / APPLICATION

(NOT USED)

3.04 TOLERANCES

(NOT USED)

3.05 CLEANING

(NOT USED)

3.06 PROTECTION

- A. Package and store materials in a manner that will prevent surface damage or contamination, distortion, breakage or structural weakening.
- B. Replace any materials damage during manufacture, shipping, storage, or erection.
- C. Protection materials in place from contamination and damage.
- D. The contractor shall submit a preliminary plan on the method for the protection of the surface of the glass balustrade/metal railing in particular protection against typhoons, cement droppings, etc. during delivery prior to installation and during & after erection, with description as to when the protection can be removed.
- E. This plan shall be acceptable to the Architect.
- F. The plan of protection shall include allowance for all installed material to be suitably protected by encasing, covering up, etc., all proposed protective measures for all components, accessories, glass, etc. as appropriate against damage due to building operations (such as covering up with adhesive tape, foam casing, or plastic sheeting the components after installation to prevent from corrosion by cement slurry from building operations above or falling objects, etc.), adverse weather or other causes.
- G. The plan of protection shall include screening the window wall and glass balustrade & railing from weld splatter, spray-on fireproofing, concrete, alkaline masonry washes, paints, and other deleterious substances. Any such soiling shall be promptly and completely removed. The design of protective screening shall be such as to provide adequate ventilation of the space between the glass and the protective screen and not induce thermal stresses in the glass. In no case shall the protective screening be placed in contact with the glass.

END OF SECTION